



From Shanghai to the Shore: The Silent Threat in Global Shipping

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Agenda

- # whoami's
- US Coast Guard & Cyber 101
- Cranes Manufactured in China
 - Why they're important.
 - What they are.
 - What we've done to protect them.

whoami's



Kenny



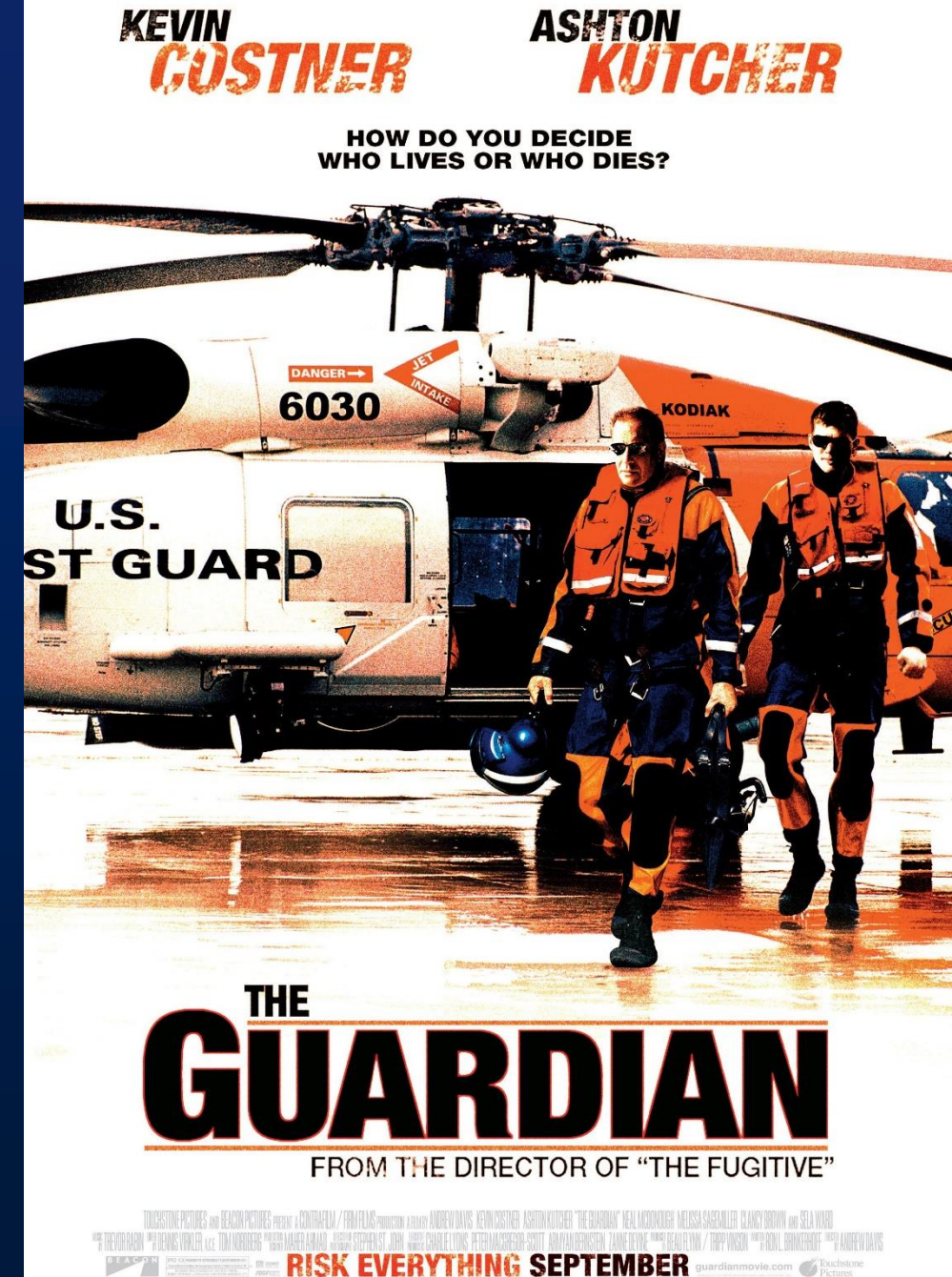
Nick



The U.S. Coast Guard does cyber?



Source: [The Claw of Knowledge](#)



U.S. Coast Guard Overview

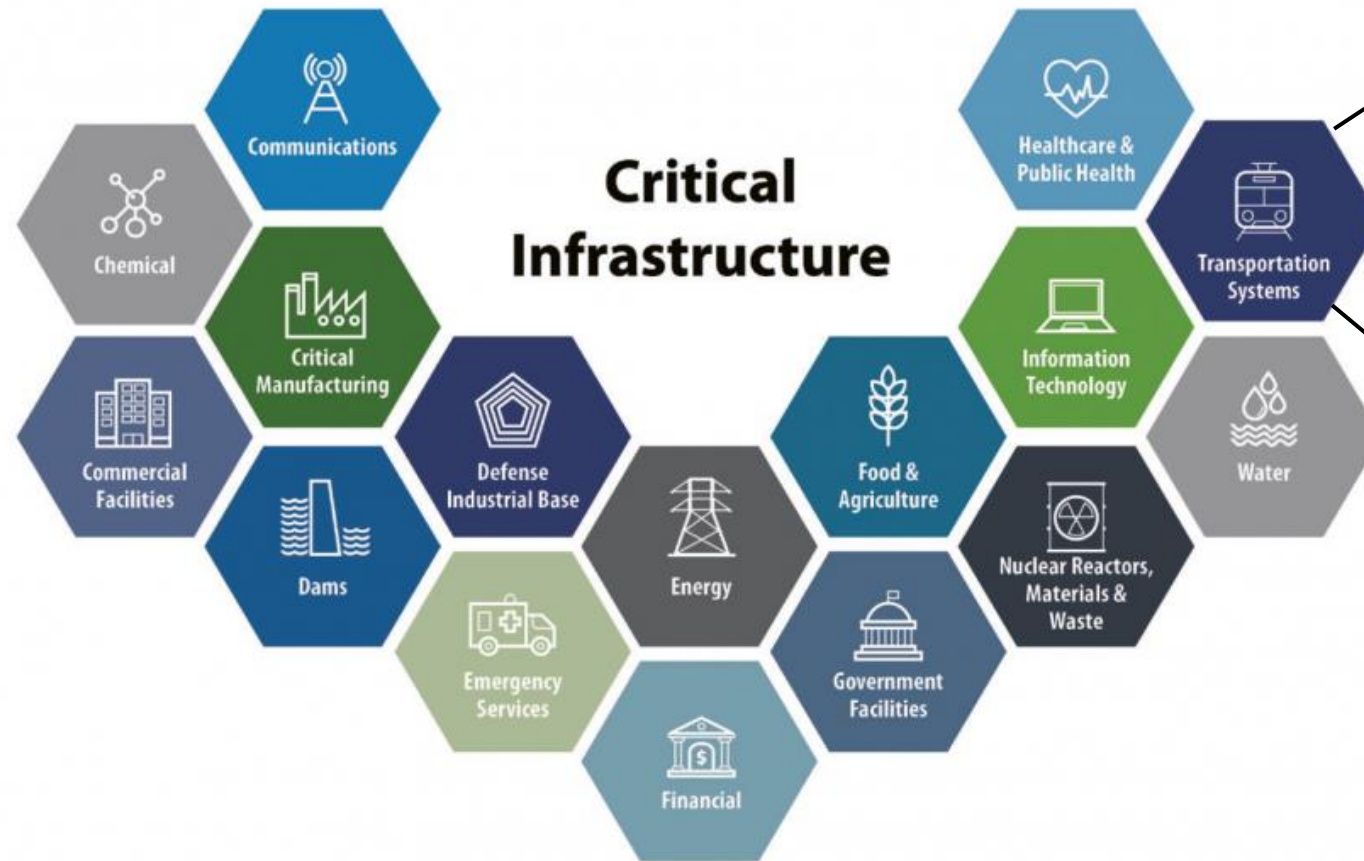


And we do cyber!

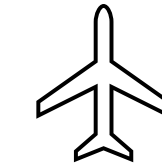


Source: [The Claw of Knowledge](#)

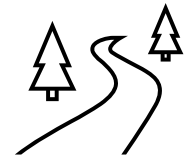
Protecting National Critical Infrastructure



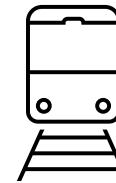
© US Cybersecurity & Infrastructure Security Agency (CISA)



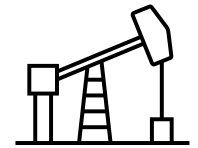
Aviation



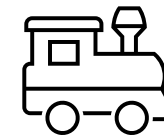
Highway and Motor Carrier



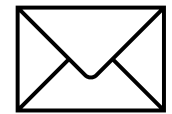
Mass Transit and Passenger Rail



Pipeline Systems



Freight Rail



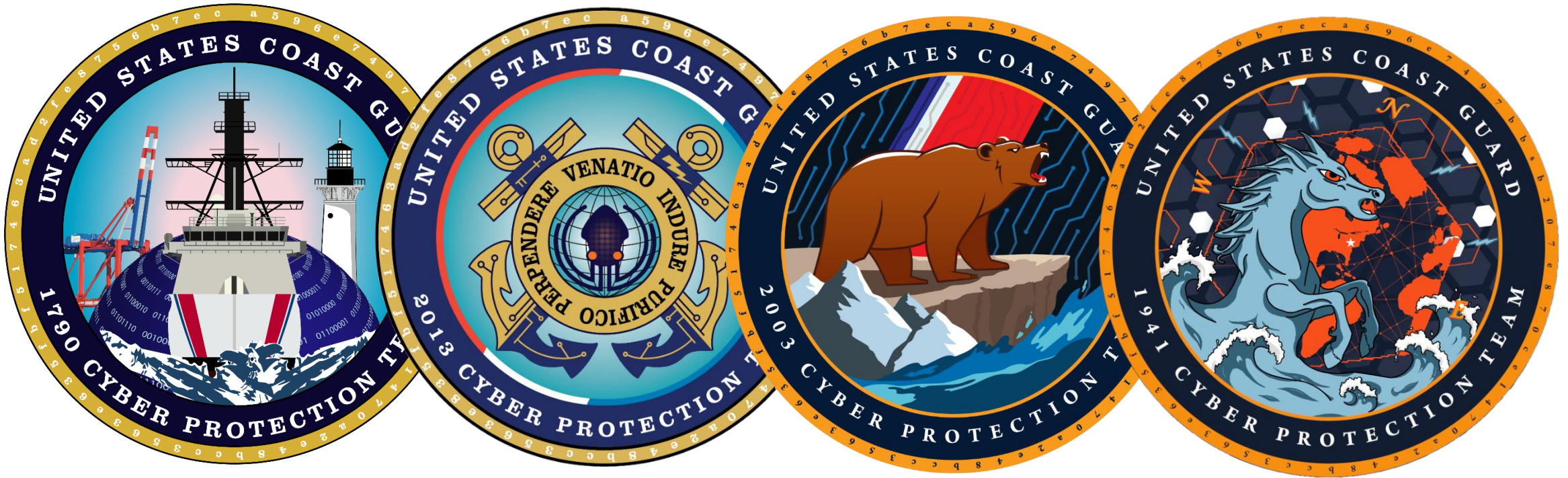
Postal and Shipping



Marine Transportation System (MTS)



U.S. Coast Guard Cyber Protection Teams



Reserve



Theory of the Case

Ship-to-shore (STS) cranes manufactured in China present a risk to the Marine Transportation System (MTS).

- The U.S. is dependent on cranes manufactured by a Chinese state-owned enterprise which could present a significant supply chain risk.
- Extensive analysis conducted by Coast Guard Cyber Protection Teams (CPTs) has revealed vulnerabilities that may enable a malicious cyber actor the ability to disrupt port operations.



Daily **Mail**

Chinese 'spy cranes' in US shipping ports ARE equipped with secret modems that could be used for espionage or sabotage, probe finds

CNN politics

Congressional probe finds communications gear in Chinese cranes, raising spying concerns

THE WALL STREET JOURNAL.

Espionage Probe Finds Communications Device on Chinese Cargo Cranes

Blaze media

Communication devices found on Chinese-built cranes located at US ports spark espionage concerns

THE WALL STREET JOURNAL.

Chinese Cargo Cranes at U.S. Ports Pose Espionage Risk, Probe Finds

Chinese manufacturer ZPMC has pressured American ports for remote access to its machines, a congressional report says

Washington Examiner

House investigation finds Chinese-made cranes at US ports have communications gear installed

FOX
BUSINESS

Chinese crane firm denies posing security risk at US ports amid investigation

Chinese Manufactured Cranes

NEW

Suspicious tech found in Chinese-made cargo cranes, fueling spying worries: Congress

BUSINESS INSIDER

China could be spying on US ports using secret tech built into cranes

The Maritime Executive
INTELLECTUAL CAPITAL FOR LEADERS

Chinese-Built Port Cranes May Be Able to Call Home On Their Own

NEWSMAX

House Committees Probe Possible Chinese Spying at US Ports

DAILY CALLER

'Clearly Overlooked This': Probe Finds Strange Communication Devices On Chinese Cranes In US Ports

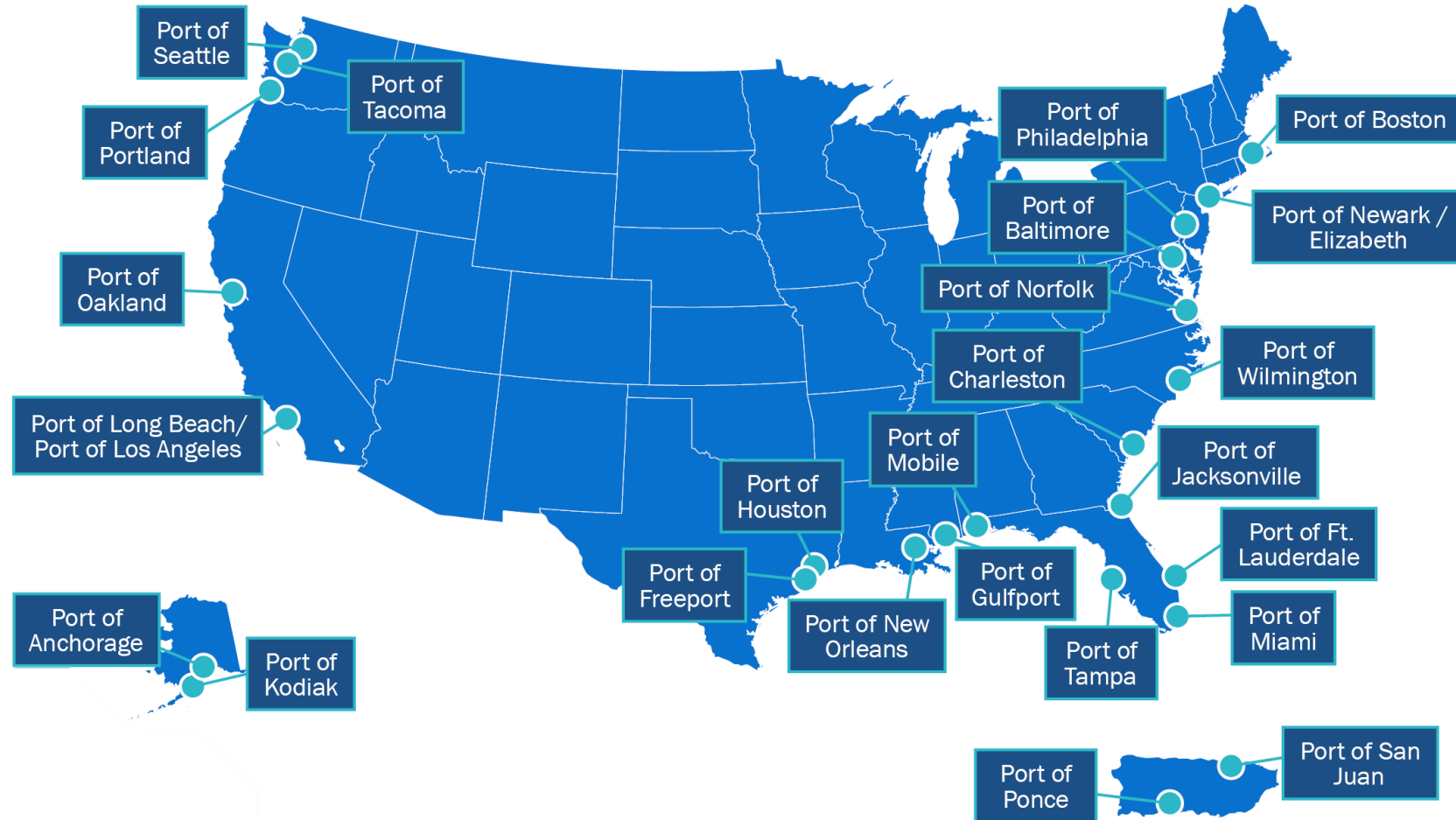


A photograph of a port worker in an orange jacket and dark cap, seen from behind, looking out at a large gantry crane at the Port of Oakland. The crane is white and has "PORT OF OAKLAND" written on its side. In the background, a large container ship is docked at the pier. The scene is set against a clear blue sky and the water of the port.

Why cranes?

- 70% of non-bulk cargo is moved by containers.
- Over 70% of the world's ship to shore cranes are made by Shanghai Zhenhua Heavy Industries (ZPMC), a Chinese State-Owned Enterprise (SOE).
- ZPMC accounts for over 80% of STS cranes in the U.S.
- Congressional interest:
 - [March 2024](#)
 - [September 2024](#)

ZPMC Cranes in the US



Supply Chain Risks

- Chinese State-Owned Enterprise (SOEs)
- Previous Supply Chain Attacks
 - Solarwinds (2020)
 - Cisco Equipment (2014-2022)
- Huawei & Salt Typhoon
- Potential impact to cranes

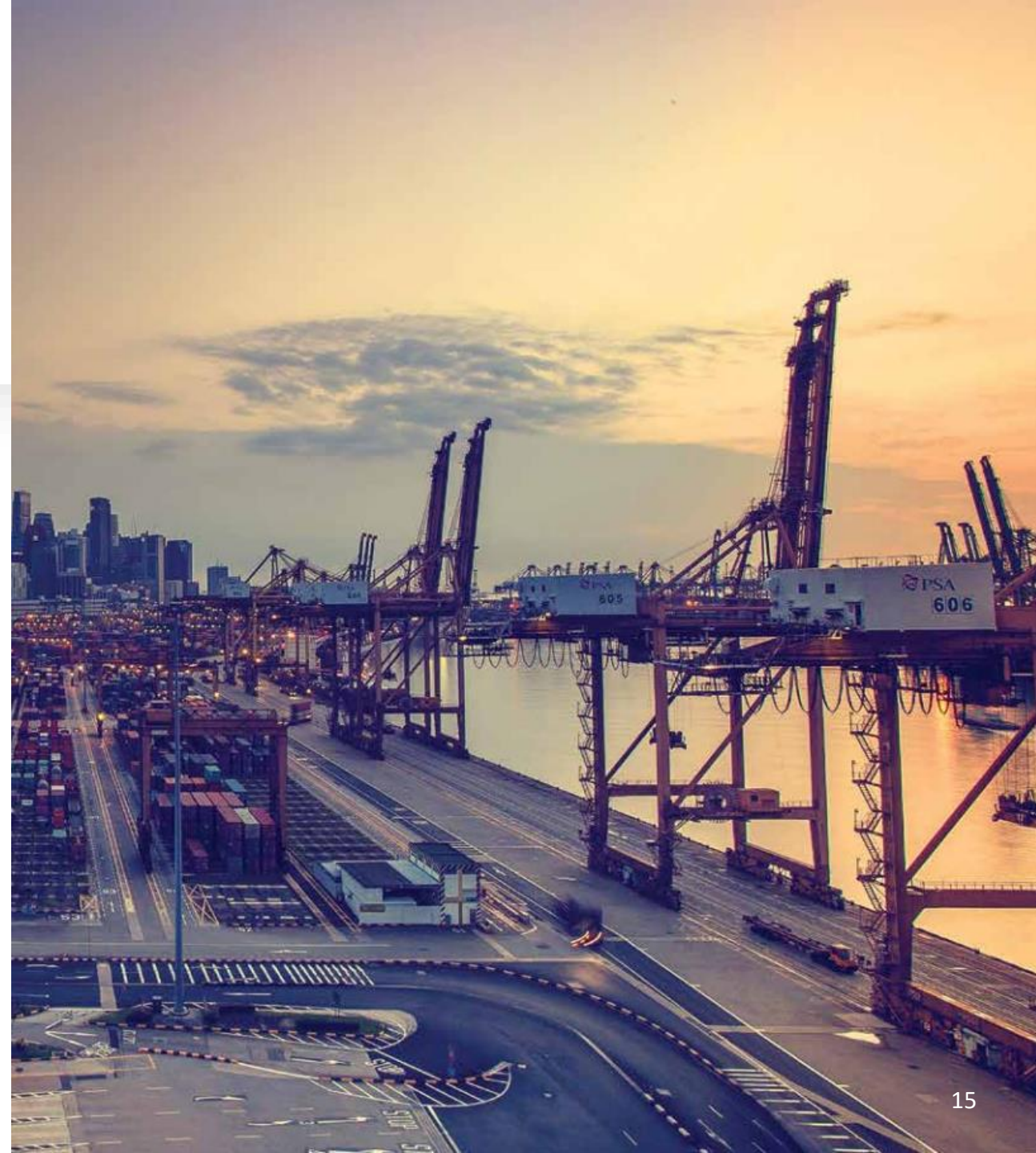


Ok, so what are we doing about it?

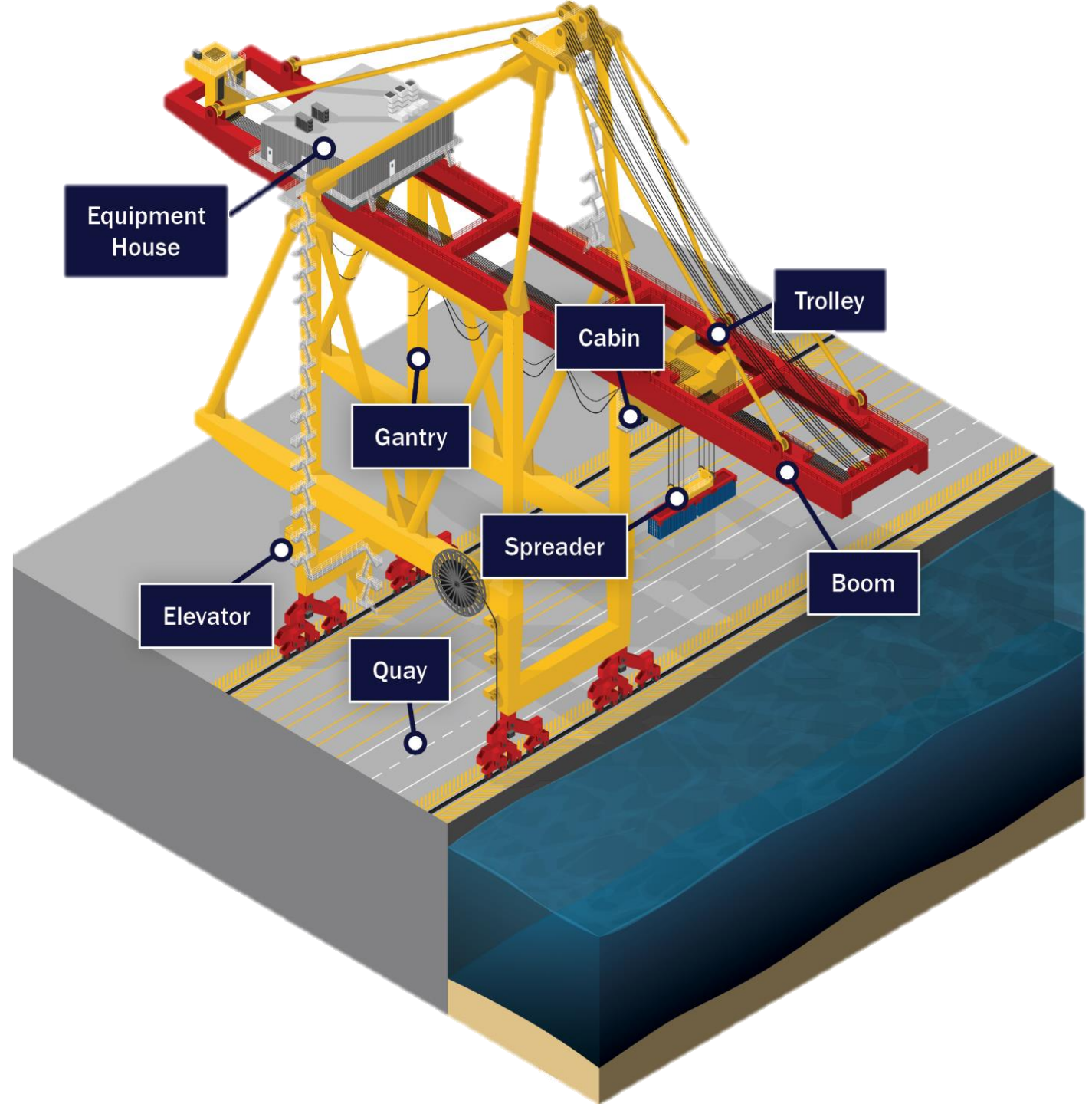


USCG Cyber Protection Team Chinese Crane Missions

- Hundreds of days sensed on cranes across multiple ports
- More time spent analyzing crane data than any other federal agency.



Parts of an STS Crane

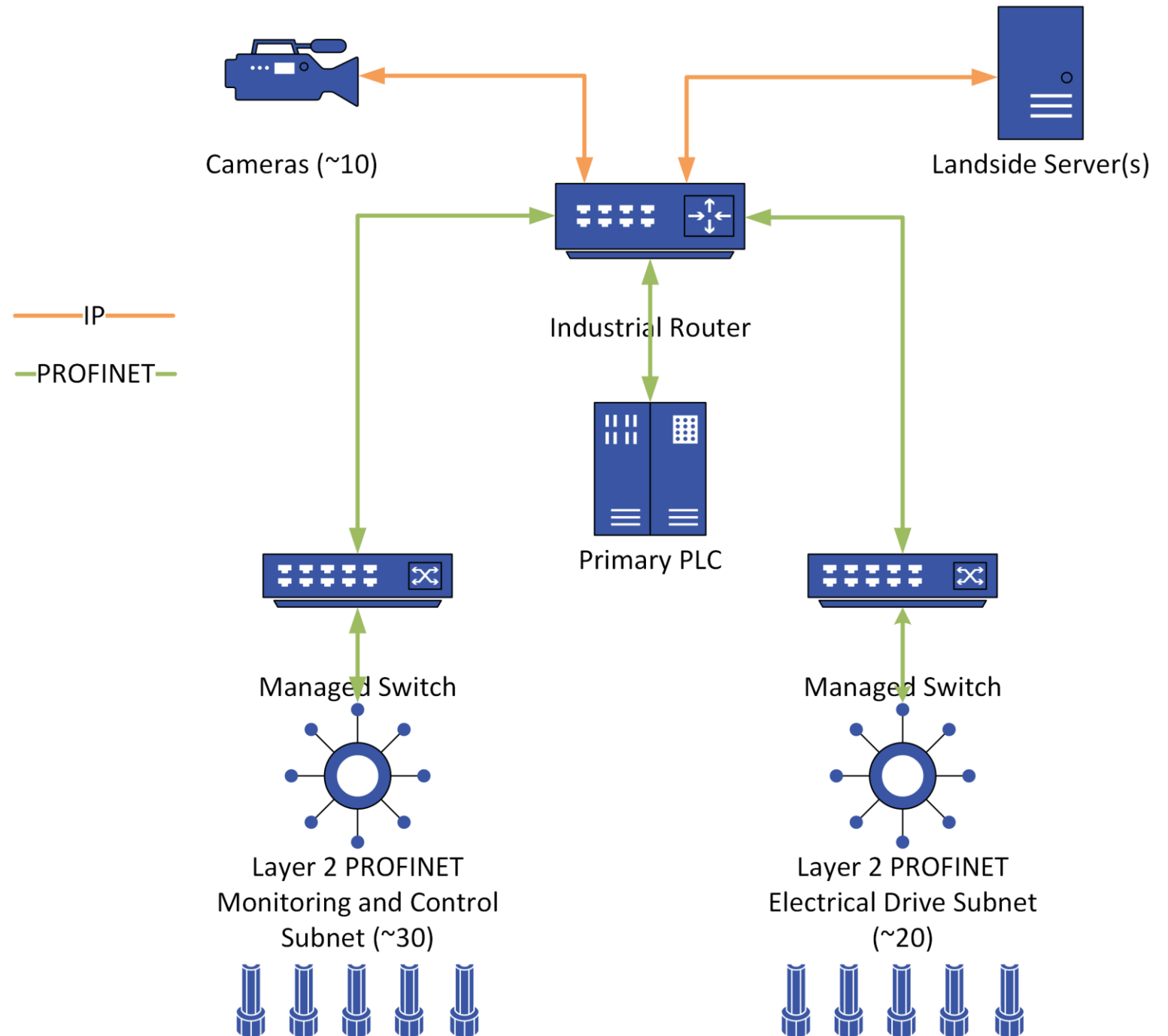


Modern Crane Features

- Features:
 - Assisted Control Systems
 - Optical Character Recognition
 - Remote Control
 - Autonomous Cranes
- Protocols:
 - PROFINET
 - Link Layer Discovery Protocol (LLDP)



Logical Structure of a Crane



PROFINET

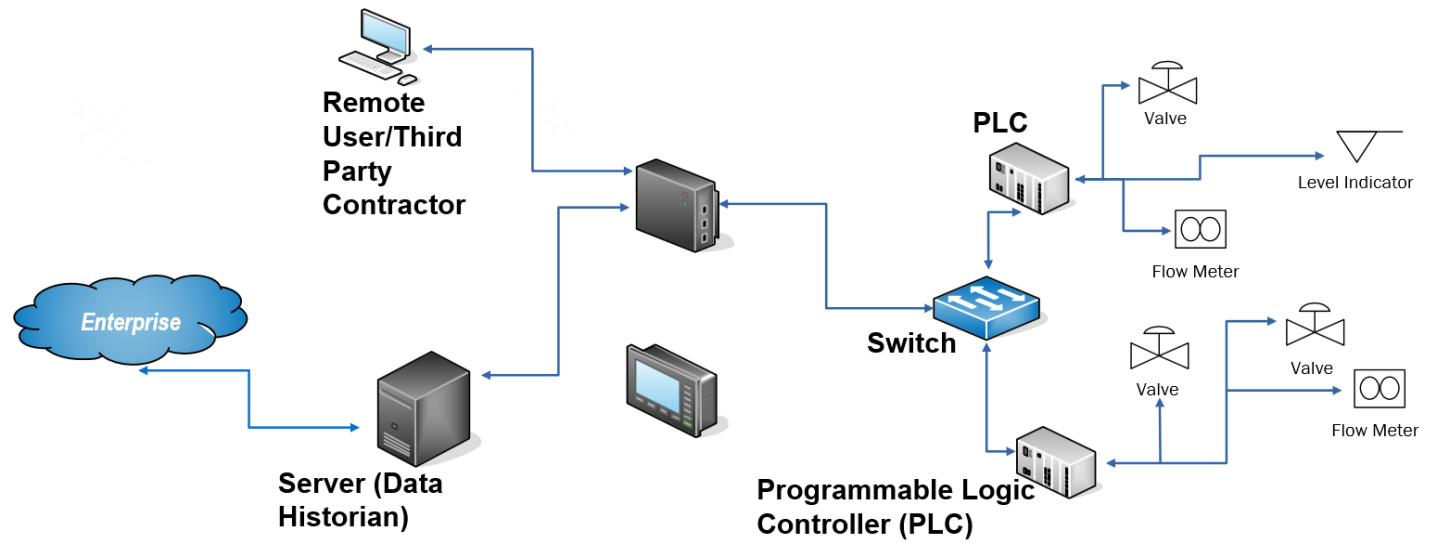
```
> Frame 12: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0
Ethernet II, Src: ABBAutom_05:29:52 (00:24:59:05:29:52), Dst: ABBAutom_0e:57:bc (00:24:59:0e:57:bc)
  Destination: ABBAutom_0e:57:bc (00:24:59:0e:57:bc)
    Address: ABBAutom_0e:57:bc (00:24:59:0e:57:bc)
      ....0... = LG bit: Globally unique address (factory default)
      ....0... = IG bit: Individual address (unicast)
  Source: ABBAutom_05:29:52 (00:24:59:05:29:52)
    Address: ABBAutom_05:29:52 (00:24:59:05:29:52)
      ....0... = LG bit: Globally unique address (factory default)
      ....0... = IG bit: Individual address (unicast)
  Type: PROFINET (0x8892)
PROFINET cyclic Real-Time, RTC1(legacy), ID:0xc000, Len: 40, Cycle: 2048 (Valid,Primary,Ok,Run)
  FrameID: 0xc000 (0xc000-0xf7ff: Real-Time(class=1 unicast): Cyclic)
  CycleCounter: 2048
  DataStatus: 0x35 (Frame: Valid and Primary, Provider: Ok and Run)
    0... = Ignore (1:Ignore/0:Evaluate): 0x0
    .0.. = Reserved_2 (should be zero): 0x0
    ..1. = StationProblemIndicator (1:Ok/0:Problem): 0x1
    ...1 = ProviderState (1:Run/0:Stop): 0x1
    ....0... = Reserved_1 (should be zero): 0x0
    ....1.. = DataValid (1:Valid/0:Invalid): 0x1
    ....0.. = Redundancy: Redundancy has no meaning for OutputCRs / One primary AR of a given AR-set is present
    ....1.. = State (1:Primary/0:Backup): 0x1
  TransferStatus: 0x00 (OK)
PROFINET IO Cyclic Service Data Unit: 40 bytes
```

0000	00 24 59 0e 3e d8	00 24 59 05 29 52	88 92 c0 00
0010	00 00 00 00 00 00	00 00 00 00 00 00	00 00 00 00
0020	00 00 00 00 80 cc 80	00 a0 78 71 63 80	80 80 80 80
0030	80 80 80 80 80 00	00 00 00 00	08 00 35 00

Wireshark view of a PROFINET frame from a ZPMC Crane

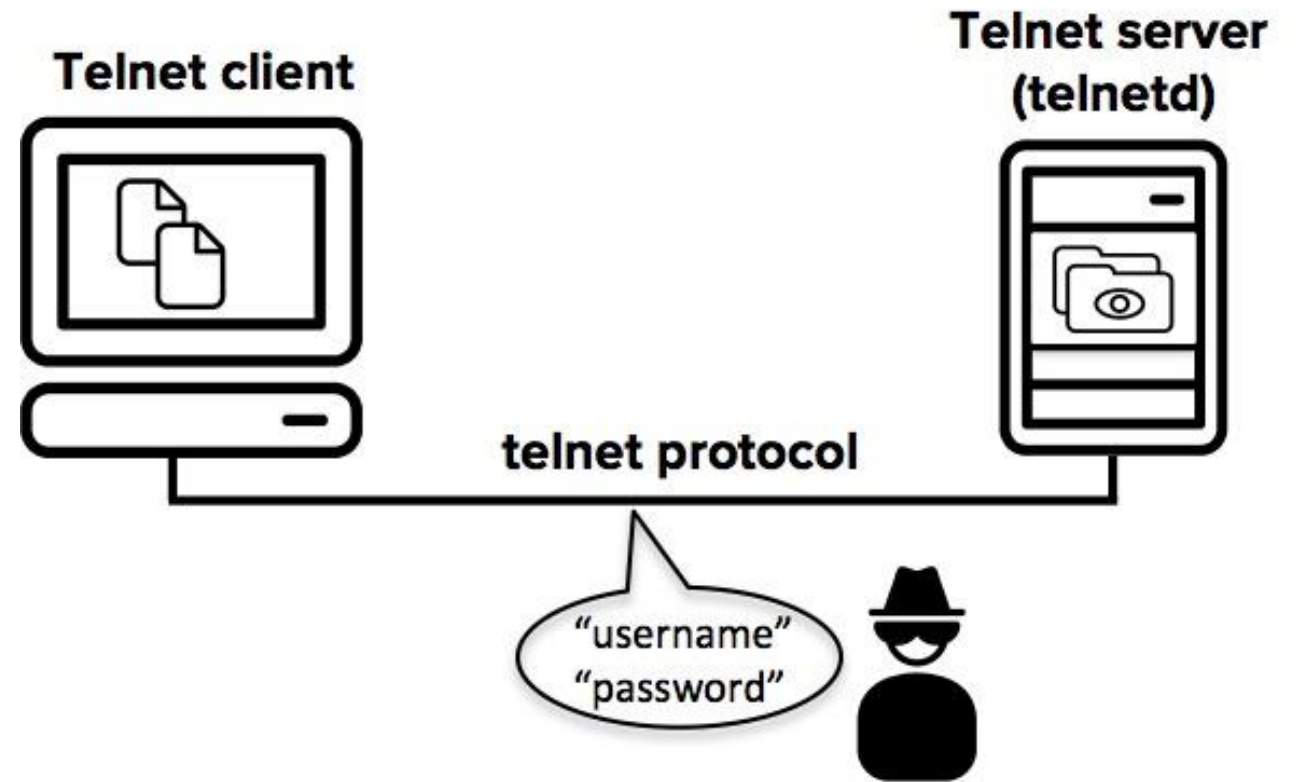
Most Common Findings

Improper network segmentation



Legacy Protocols

- Telnet
- SMBv1
- LLMNR



Source: [Countering Password Stealing Attacks - Replace telnet with SSH.](#)

End-of-Life OS



Shared Accounts

Creds to Shared Account



Daily **Mail**

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Espionage Probe Finds Communications Device on Chinese Cargo Cranes

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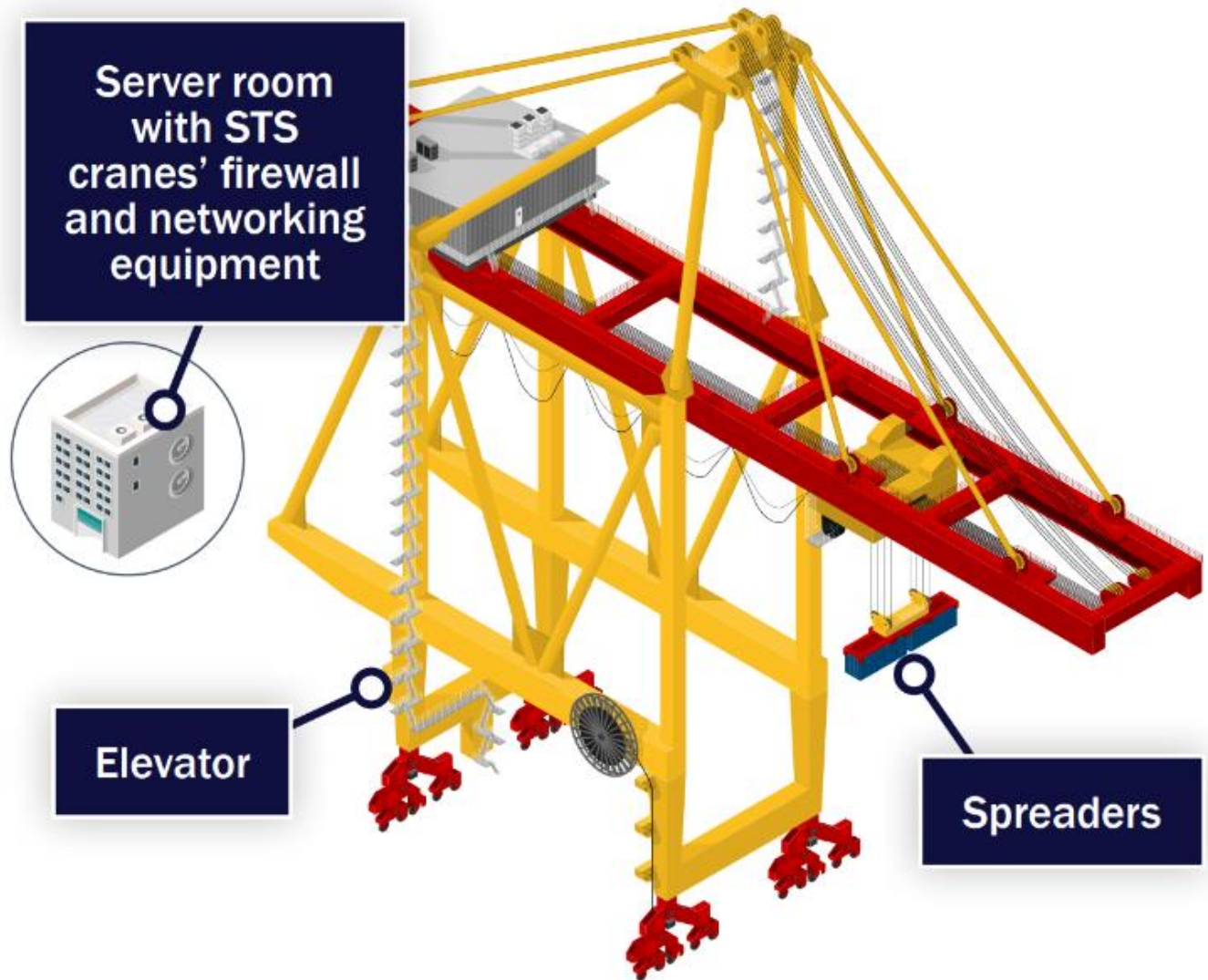
House investigation finds Chinese-made cranes at US ports have communications gear installed

Cellular Modems

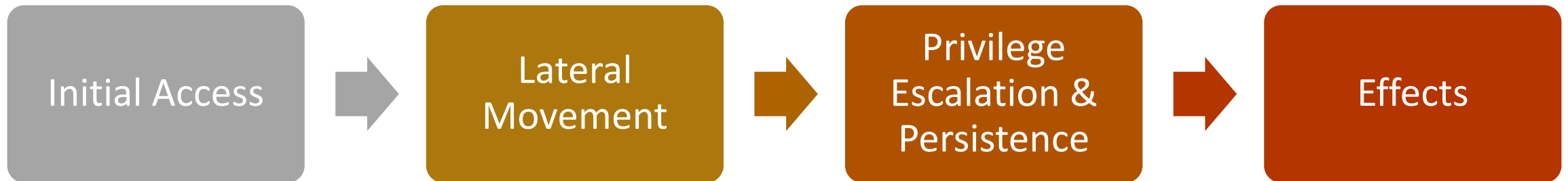
This was us!



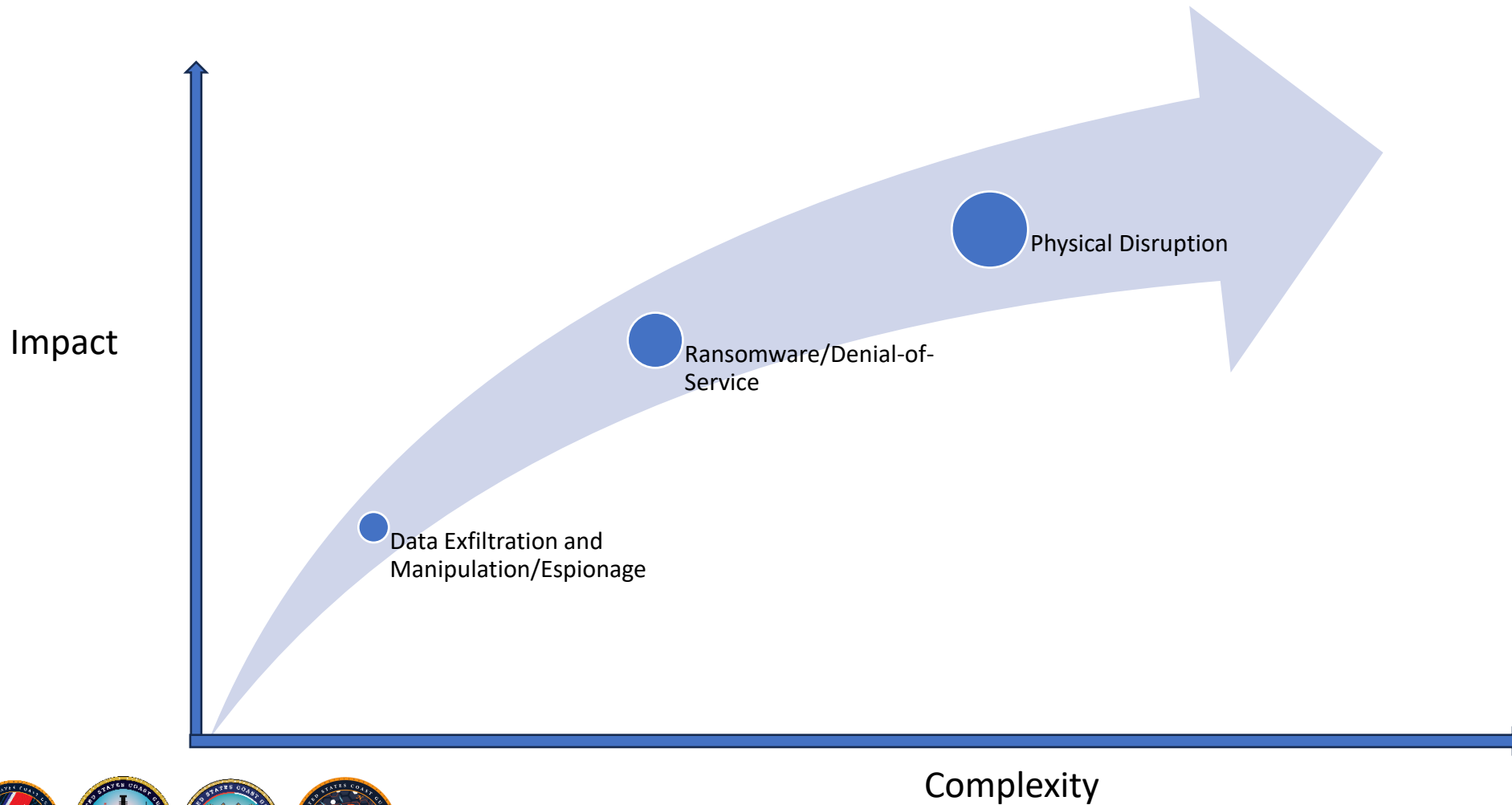
Where we
found
cellular
modems.



Attack Paths



Potential Cyber Effects





Malicious Cyber Activity

Source: AI Generated

Best Practices

- Scrutinize contract language.
- Restrict remote access.
- Standard enterprise cybersecurity best practices!





Thank you!

U. S. COAST GUARD

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